

Minne"Soda"

An Analysis of Convenience Store Data &
Subgroup Trends on Heath

Sophie Scherkenbach and Megan Halpin

Research Question

What is the health quality of people, and does it vary by demographic factors such as age, sex, education level, etc.?

Our Data

We used a data set that recorded health scores - HEI Scores - of Minnesotans over different time periods to see if health initiatives worked

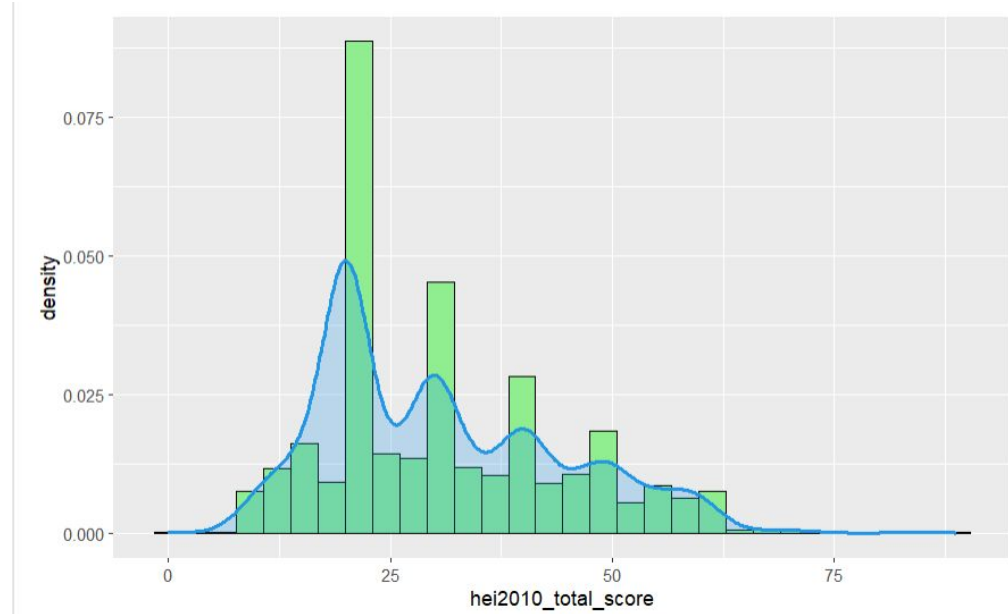
Variables of interest: HEI Score, sex, education level, shopping frequency, HEI over time

Results

Density Curve of HEI Scores

We notice a right skew,
indicating more people had
lower values

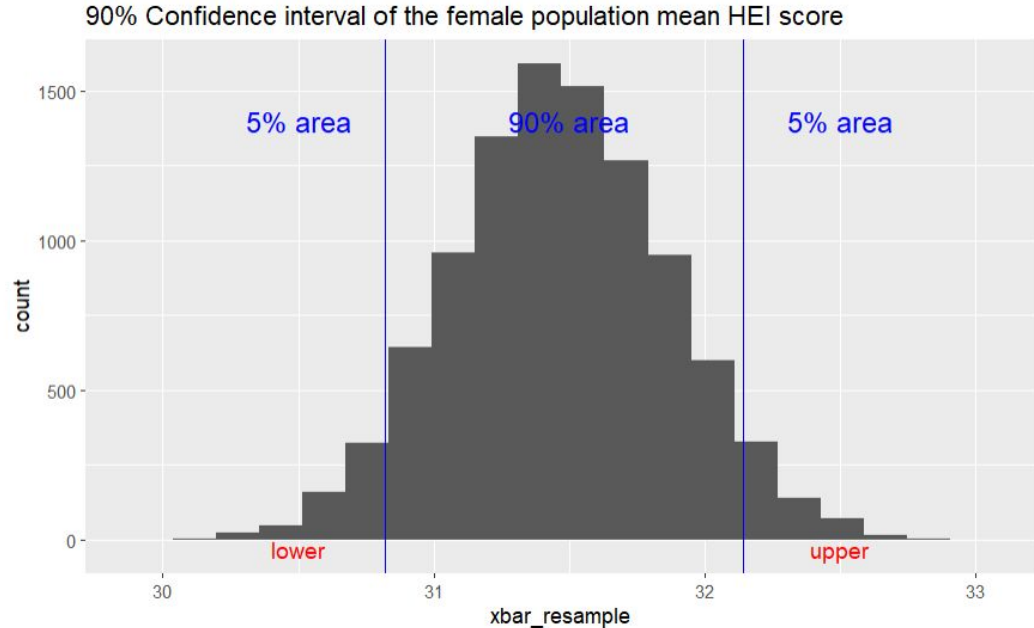
Center is around 25 - 30,
indicating poor diet health



Female Mean Distribution of HEI Score

We are 90% confident the mean HEI score for women is between 30.82 and 32.14 and 32.14

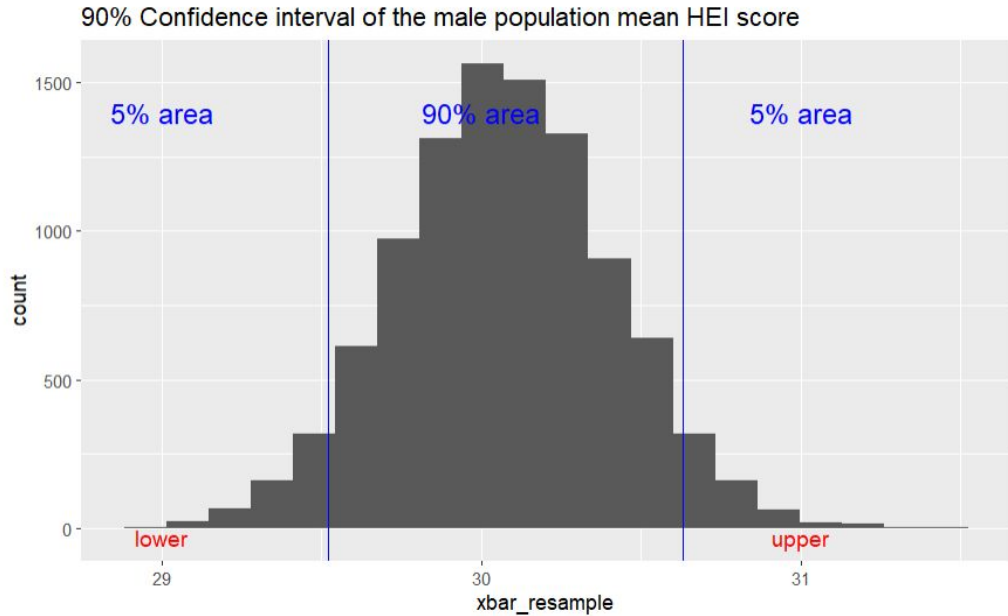
Higher than the total population



Male Mean Distribution of HEI Score

We are 90% confident the mean HEI score for men is between 29.52 and 30.63

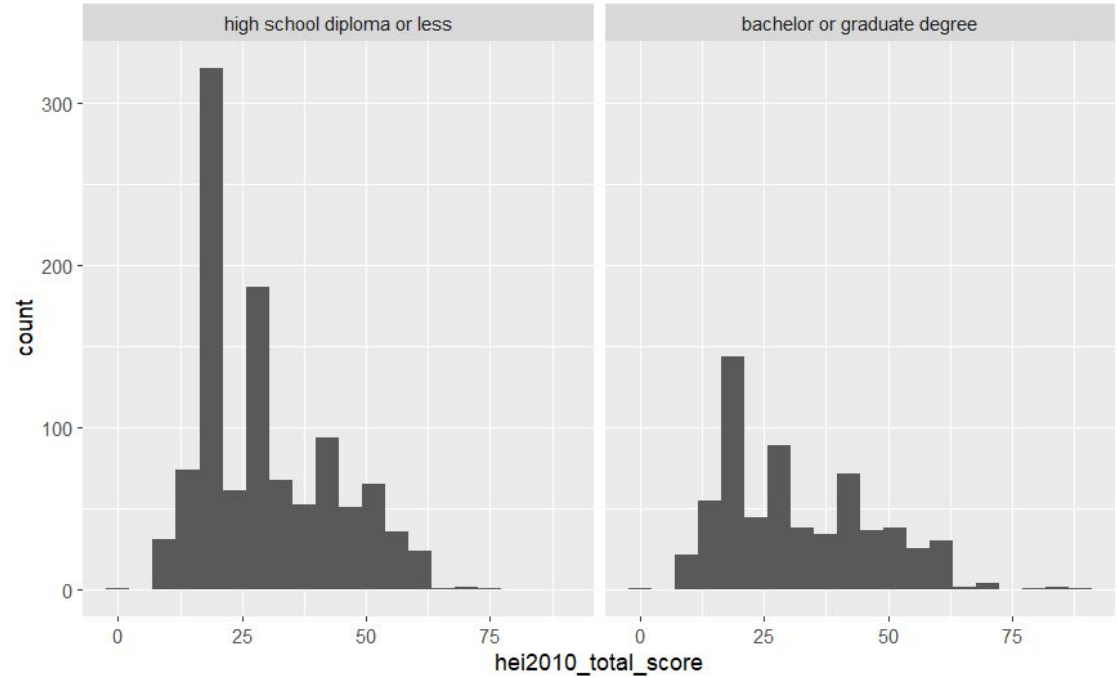
Lower than the total population as well as the female population



Mean HEI Score in different education levels

P value: 0.006

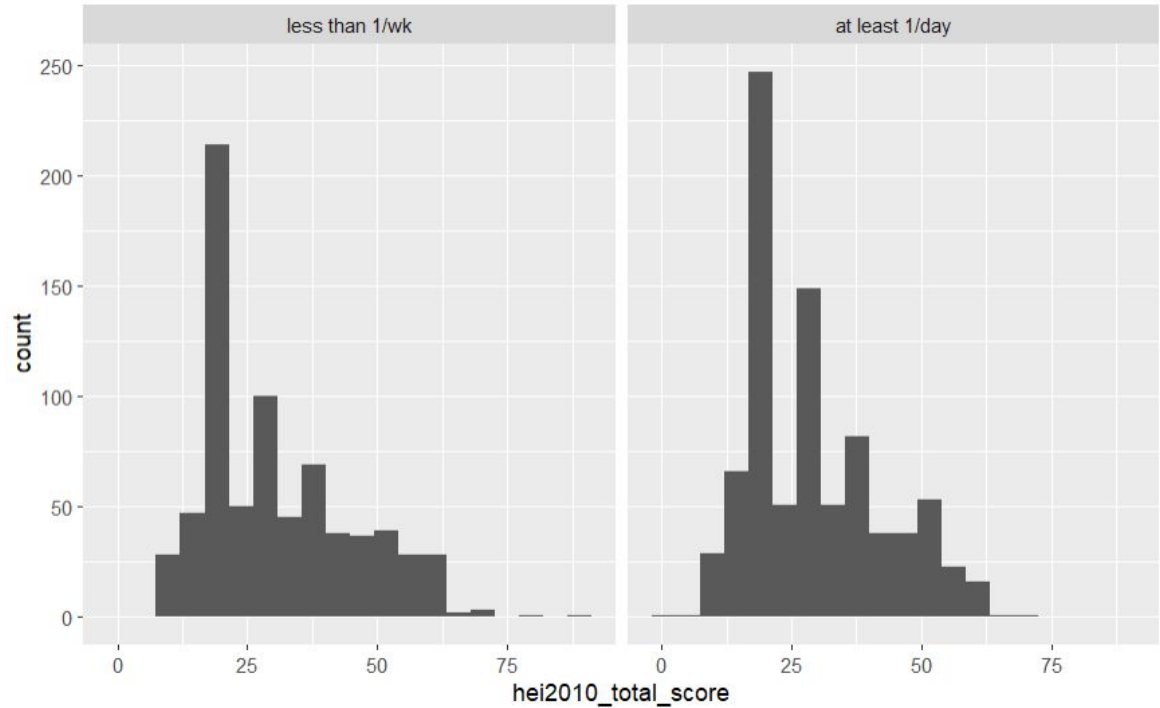
The mean score of people with college degrees is higher than the mean score of high school diploma or lower



Mean HEI Score in different shopping frequency

P value: 0.027

The mean score of people who shop less than once a week is higher than those who shop daily



Chi Squared Table of Time and HEI Score

P value: 0.077

The p value suggests that we fail to reject the null hypothesis which means that there is likely no statistically significant difference in HEI scores over time

Pearson's Chi-squared test

```
data: table_hei_time  
X-squared = 15.509, df = 9, p-value = 0.07788
```

Conclusion

Based on tests ran, HEI score in Minnesota is bad

Women, those who highly educated, and people who only shop once a week have higher HEI scores on average, though the scores were still poor

According to the USDA website, mean HEI scores for the whole country was around 60, meaning Minnesotians have poor health comparatively

Health initiatives put in place by the Minnesota government were well founded, and further intervention may be required